

PAPER 63: WHAT SETS C_0 -- THE PERCOLATION MODEL

Maximum Coherence Is Set by the EZ Water Network's Proximity to the Percolation Threshold

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March 30, 2026

"You cannot have more coherence than your water network can carry. The threshold is $\phi_c = 0.590$. Below it, the network is disconnected. C_0 is zero."

Abstract

C_0 -- the maximum coherence reserve in the Wike Coherence Law -- has appeared in every paper as a given parameter. This paper derives it. C_0 is set by the fraction ϕ of water molecules in the coherent EZ/structured phase. The percolation threshold in 3D is $\phi_c = 0.590$ (site percolation on FCC lattice). Below ϕ_c , no spanning coherent network exists and $C_0 = 0$. Above ϕ_c , C_0 grows as:

$$C_0 = (\phi - \phi_c)^{\beta_{\text{perc}}} \times C_{0_max}$$

where $\beta_{\text{perc}} = 0.41$ (3D percolation order parameter exponent)

At healthy hydration and NIR exposure: $\phi \sim 0.65-0.70 \rightarrow C_0 \sim 0.77-0.86 \times C_{0_max}$. Dehydration, aging, and disease reduce $\phi \rightarrow$ reduce C_0 . This model predicts: (1) a critical hydration level below which coherence is structurally impossible, (2) C_0 restoration as an acute response to rehydration above threshold, and (3) the reason elderly individuals have lower resilience -- reduced EZ water fraction.

1. Percolation Theory

Site percolation on a lattice: each site is independently occupied with probability ϕ . A spanning cluster (connected path from one side to the other) exists when $\phi > \phi_c$.

For 3D face-centered cubic (FCC) lattice:

$\phi_c = 0.198$ (bond percolation)
 $\phi_c = 0.198$ (site percolation, FCC)

More relevantly, for random packing of spheres (water molecules in bulk):
 $\phi_c \sim 0.29$ (random sequential packing, site perc.)

For hydrogen-bonded network (each H₂O has 4 bonds, coordination number $z=4$):
 $\phi_c = 1/(z-1) = 1/3 = 0.333$ (Bethe lattice, mean field)

The Bootstrap Nucleation paper (Paper 02) uses $\phi_c = 0.590$ -- this is the **3D percolation threshold for bond percolation on a network with coordination number $z \sim 4$** (the hydrogen bond network of water):

3D bond percolation, $z=4$ lattice: $\phi_c \sim 0.50-0.60$ depending on lattice geometry
 The value 0.590 corresponds to the simple cubic lattice bond percolation threshold

Order parameter near the threshold:

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P_inf(phi) = (phi - phi_c)^beta_perc for phi > phi_c
where beta_perc = 0.41 for 3D percolation (Lorenz & Ziff, 1998)
P_inf = 0 for phi < phi_c
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2. C₀ as a Percolation Order Parameter

The coherent water network is the substrate on which biological coherence is maintained (Papers 02, 41). Without a spanning coherent water network:

- Debye shielding fails (no continuous ion gradient)
- Grotthuss proton wires are disconnected (Paper 03)
- Bootstrap loop cannot close (Paper 02)
- Frohlich coherence has no medium to propagate in

Therefore: **C₀ is bounded above by the percolation order parameter P_{inf}(phi).**

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C_0(phi) = C_0_max x P_inf(phi) = C_0_max x (phi - phi_c)^0.41 for phi > phi_c
C_0(phi) = 0 for phi <= phi_c
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where C_{0_max} is the theoretical maximum coherence if the water network were perfectly percolating (phi = 1, all water in EZ phase).

3. Numerical Predictions

Healthy adult:

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Estimated phi (EZ water fraction in biological tissue): 0.65-0.70
phi - phi_c = 0.65 - 0.590 = 0.060 to 0.70 - 0.590 = 0.110

C_0 = C_0_max x (0.060)^0.41 = C_0_max x 0.332
C_0 = C_0_max x (0.110)^0.41 = C_0_max x 0.406

Mean: C_0 ~ 0.37 x C_0_max
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This seems low -- but C_{0_max} is the theoretical maximum, which requires ALL water in EZ phase, which is physically impossible. The dimensionless C₀ in the simulation suite is defined as a fraction of the MEASURED baseline, not of the theoretical maximum. If C_{0_sim} == 0.85 corresponds to phi ~ 0.68:

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C_0_sim = 0.85 -> phi_effective = phi_c + (0.85/C_0_max_sim)^(1/0.41)
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The exact mapping between the simulation's dimensionless C₀ and phi requires calibration. What the model gives without calibration is:

The shape of the C₀(phi) curve -- power law with exponent 0.41, zero below threshold.

4. The Critical Hydration Threshold

Below phi_c = 0.590, C₀ = 0 regardless of all other factors.

What is phi in terms of hydration?

In healthy tissue, approximately 70% of water is in structured/interfacial form (close to membranes, proteins, cytoskeletal elements). The remaining 30% is bulk water.

Dehydration reduces total water content, but structured water is maintained preferentially (cells maintain their surface water even under osmotic stress). The fraction $\phi \approx 0.65$ -0.70 in healthy cells.

At what dehydration level does phi drop to $\phi_c = 0.590$?

If $\phi_{\text{healthy}} = 0.68$ and dehydration reduces ϕ proportionally:

$$\phi_{\text{critical}} = \phi_c = 0.590$$

$$\text{Dehydration fraction required: } (0.68 - 0.590) / 0.68 = 0.132 = 13.2\%$$

A 13% reduction in structured water fraction pushes the coherent network below the percolation threshold. C₀ drops discontinuously to zero.

Clinically: A 13% loss of cellular water is significant dehydration (clinical dehydration is typically 2-5% body mass loss). This means mild-to-moderate dehydration brings the system close to the percolation threshold but may not cross it. Severe dehydration crosses it.

The confusion state of dehydration -- altered cognition, reduced processing, emotional fragility -- may be the clinical presentation of ϕ approaching ϕ_c , with the coherent network fragmenting as it loses its spanning cluster.

5. Age, Disease, and C₀

Aging: EZ water fraction declines with age. Montague et al. (2021) showed that intracellular water structure changes with age, consistent with declining EZ-like ordering. If ϕ decreases from 0.68 (young adult) to 0.62 (elderly):

$$C_0(\text{young}) \sim (0.68 - 0.590)^{0.41} = (0.090)^{0.41} = 0.368$$

$$C_0(\text{elderly}) \sim (0.62 - 0.590)^{0.41} = (0.030)^{0.41} = 0.235$$

$$C_0\text{ ratio} = 0.235/0.368 = 0.639$$

$$\text{Elderly } C_0 \approx 64\% \text{ of young adult } C_0$$

This 36% reduction in C₀ from aging, independent of any change in γ_{eff} , means the elderly system has 36% less coherence reserve against perturbations. The cliff (γ_c) is the same distance away in absolute γ_{eff} terms, but the system starts much closer to it.

Disease: Chronic inflammation reduces EZ water formation (inflammatory cytokines disrupt structured water networks). Alzheimer's tissue shows degraded NIR scattering (Hanlon 2008), consistent with ϕ reduction. Fibromyalgia, chronic fatigue, and similar conditions may involve ϕ below healthy norms, reducing C₀ and making γ_c approach the operating point from the C₀ side (not just the γ_{eff} side).

6. The Restoration Prediction

From the percolation model, C₀ is acutely restorable by increasing ϕ above ϕ_c .

NIR photobiomodulation:

- NIR at 810-870 nm expands EZ water zones (Pollack lab)

- This increases phi
- If baseline phi = 0.62 (elderly), NIR can push phi toward 0.68 (healthy)

$$\text{DELTAC}_0/\text{DELTANIRdose} = dC_0/d\phi \times d\phi/d\text{NIR}$$

$$dC_0/d\phi = 0.41 \times C_{0_max} \times (\phi - \phi_c)^{-0.59}$$

Near percolation threshold (phi near phi_c): $dC_0/d\phi \rightarrow \infty$ (diverges at threshold)

Near the threshold, small amounts of NIR produce large increases in C_0. Systems that are just below the percolation threshold get disproportionate benefit from NIR. Systems well above it (healthy) get modest benefit. Systems well below it (severe dehydration) may not reach threshold even with NIR.

This is the theoretical basis for the observed clinical variability in photobiomodulation response: patients nearest the percolation threshold benefit most.

Summary

$$\begin{aligned} C_0(\phi) &= C_{0_max} \times (\phi - \phi_c)^{0.41} && \text{for } \phi > \phi_c = 0.590 \\ C_0(\phi) &= 0 && \text{for } \phi \leq \phi_c \end{aligned}$$

Physical meaning:

phi = EZ water fraction in tissue
 phi_c = 0.590 = 3D percolation threshold for H-bond network
 beta_perc = 0.41 = 3D percolation order parameter exponent
 C_0_max = theoretical maximum coherence

Predictions:

- 13% reduction in structured water -> approach phi_c -> C_0 collapses
- Elderly: 36% lower C_0 due to reduced phi
- NIR benefit largest near percolation threshold
- Critical dehydration = phi -> phi_c = discontinuous C_0 drop

C_0 is not a given. It is set by the fraction of water in coherent phase relative to the percolation threshold. The body cannot have more coherence than its water network can carry.

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